Supplementary Material

Artificial Intelligence for Angiographic Sequence Analysis: A Systematic Review of Temporal Modeling, Clinical Readiness, and Research Gaps

Table S1: Title/abstract screening validation sample: all 134 records the pipeline labelled INCLUDE (census) and 357 randomly sampled records it labelled EXCLUDE (see Methods, main text). **R1** and **R2**: independent decisions of the two human reviewers (I = include, E = exclude). **Final**: consensus label after discussion († marks the four records on which the reviewers initially disagreed). **Pipeline**: stage-1 LLM screening decision.

Record	Title	R1	R2	Final	Pipeline
R0005	Precise Staging of Diabetic Retinopathy Through Machine Learning Analysis of Leakage Source Characteristics: A Radiomics-Driven Approach Integrating OCT-A and FFA.	E	E	E	E
R0018	A Landmark-Guided Dual-Stream Synergistic Framework for Automated Intracranial Aneurysm Detection in Magnetic Resonance Angiography.	E	E	E	E
R0025	Automated assessment of right ventricular systolic function from coronary angiograms with video-based artificial intelligence algorithms: development, validation, comparison against humans, and prospective deployment.	I	I	I	I
R0028	Coronary stenosis assessment: AI-based CT quantification, visual analysis of invasive angiography, and quantitative coronary angiography.	E	E	E	E
R0031	Five-year systemic complications in diabetic retinopathy with integrated optical coherence tomography angiography and glycated hemoglobin.	E	E	E	E
R0055	New cardiovascular risk factor and clinical surrogate: epicardial fat.	E	E	E	E
R0058	S-MEOD: A Novel Evaluation Metric for Frame-Based Medical Object Detection.	I	I	I	I
R0061	Classification of retinal diseases based on optical coherence tomography angiography using cross-modal transfer learning of domain-specific foundation AI models.	E	E	E	E
R0067	Retinal vessel imaging in inflammatory disease: From endothelial dysfunction to clinical evidence and translation.	E	E	E	E
R0079	Deep Learning-Based Augmented Contrast-Enhancement and Denoising for Reduced-Iodine and Low-Radiation 70-kVp Cerebral CT Angiography: A Prospective Study.	E	E	E	E
R0082	Deep Learning-Derived Body Composition Analysis Predicts Long-Term Mortality After Transcatheter Aortic Valve Replacement.	E	E	E	E
R0097	Tissue clock-guided prediction and intervention of futile recanalization: towards precision therapy in mechanical thrombectomy.	E	E	E	E
R0099	AngioCAD: A public x-ray angiography dataset and an adaptive fusion framework for stenosis detection.	I	I	I	I
R0100	Real-time planar angiographic cerebral perfusion imaging.	I	I	I	I
R0125	Coronary Computed Tomography and Artificial Intelligence-based Plaque Analysis.	E	E	E	E
R0164	Dual-energy CT clot and peri-clot radiomics for predicting complete reperfusion and clinical outcome following endovascular therapy in acute ischemic stroke.	E	E	E	E
R0177	Innovations In Artificial Intelligence-Guided Atherosclerosis Cardiovascular Imaging: From Detection to Prognosis.	E	E	E	E
R0209	Construction of classification model and analysis of risk factors in patients with multi-vessel coronary artery disease.	E	E	E	E

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Table S1 – *continued from previous page*

Record	Title	R1	R2	Final	Pipeline
R0222	Integrating traditional statistics and explainable machine learning for risk stratification of atrial fibrillation recurrence after pulsed field ablation: a hypothesis-generating single-center study.	E	E	E	E
R0249	Predicting the Severity of Infrapopliteal Artery Lesions in Patients With Peripheral Artery Disease Using Interpretable Machine Learning.	E	E	E	E
R0262	Machine Learning-Identified Potential Interaction Between Clazosentan and Nicardipine in Patients with Subarachnoid Hemorrhage.	E	E	E	E
R0265	AI-Based Pulmonary Embolism Detection: The Added Value of a False-Positive Reduction Module over a Region Proposal Network.	E	E	E	E
R0267	Importance of coronary artery lumen size in the relationship between coronary artery plaque and vessel-specific ischemia: A post hoc analysis of CREDENCE and PACIFIC-1.	E	E	E	E
R0300	Application of machine learning for sex-specific risk prediction of coronary artery calcification.	E	E	E	E
R0309	Deep learning for estimating right ventricular function from routine coronary angiography.	I	I	I	I
R0331	Diagnostic performance of AI-assisted intraoperative imaging for parathyroid identification: a systematic review.	E	E	E	E
R0339	Artificial intelligence risk stratification from dynamic digital subtraction angiography radiomics predicts pulmonary embolism and associates with clinical outcomes in deep vein thrombosis: A retrospective cohort study.	I	I	I	I
R0352	DOSTA-Net: Domain-Shuffle Temporal Attention Network for Vessel Extraction in X-Ray Coronary Angiography Using Synthetic Data.	I	I	I	I
R0363	MG-HGLNet: A Mixed-Grained Hierarchical Geometric-Semantic Learning Framework with Dynamic Prototypes for Coronary Artery Lesions Assessment.	E	E	E	E
R0369	CCTA as a Screening Tool (Invited Commentary).	E	E	E	E
R0376	Machine learning-enabled prediction of hemorrhagic transformation post-thrombectomy using quantitative DSA.	I	I	I	I
R0384	Association of Serum Vitamin D With Macular Microvascular Structure in Type 2 Diabetic Mellitus Without Diabetic Retinopathy: A Cross-Sectional Study.	E	E	E	E
R0394	Machine learning and SHAP values for predicting coronary artery disease risk in Xinjiang, China.	E	E	E	E
R0419	Systemic and local vascular features in branch retinal vein occlusion: analysis of the retinal age gap and crossing pattern.	E	E	E	E
R0423	Performance of deep-learning reconstruction combined with metal artifact reduction algorithm for dual-energy computed tomography angiography in intracranial aneurysm coil embolization.	E	E	E	E
R0425	Imaging-Based Artificial Intelligence in Vascular and Interventional Radiology: A Narrative Review.	E	E	E	E
R0452	Early clinical experiences with AI-based EVAR planning using the Endoleak Risk Index support its value for individualized decision-making and education.	E	E	E	E
R0467	Development and validation of a machine learning-based predictive model for coronary heart disease risk in middle-aged and young adults.	E	E	E	E
R0474	Feasibility of deep learning-based cancer detection in ultrasound microvascular images.	E	E	E	E
R0477	Deep learning-reconstructed time-maximum intensity projection versus iterative reconstruction for collateral assessment in acute anterior circulation ischemic stroke.	E	E	E	E
R0483	Artificial Intelligence-Assisted Indocyanine Green Angiography for Perforators Identification in the Anterolateral Thigh Flap.	E	E	E	E
R0488	Anatomical, physiological and inflammatory characterization of nonculprit vessels in patients undergoing primary PCI for ST-elevation myocardial infarction in the presence of multivessel disease: Rationale and design of the PICNIC study.	E	E	E	E
R0489	Automatic Image Co-registration of Carotid Angiography and Intravascular Optical Coherence Tomography Based on Machine Learning Method: A Pilot Feasibility Study.	E	E	E	I

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Table S1 – *continued from previous page*

Record	Title	R1	R2	Final	Pipeline
R0490	Comparison of precise imaging and iterative reconstruction techniques at low doses using the dose right index in 100-kVp cerebral CT angiography.	E	E	E	E
R0492	Robust Brain Extraction Tool for Nonenhanced CT and CT Angiography: CTA-BET.	E	E	E	E
R0506	Magnetic Microrobot With Drilling-Sensing Dual Functionality for Targeted Biopsy of Deep-Seated Tracheal Microlesions.	E	E	E	E
R0509	Few-shot learning for highly accelerated 3D time-of-flight MRA reconstruction.	E	E	E	E
R0514	Identifying Pathogenesis of Acute Coronary Syndromes using Sequence Contrastive Learning in Coronary Angiography.	I	I	I	I
R0521	Automated and explainable coronary angiogram interpretation for selection of coronary artery bypass grafting candidates using artificial intelligence.	I	I	I	I
R0529	Temporal consistency-aware network for renal artery segmentation in X-ray angiography.	I	I	I	I
R0533	A Dynamic Machine Learning Model to Predict Angiographic Vasospasm After Aneurysmal Subarachnoid Hemorrhage.	E	E	E	E
R0537	Predicting Significant Stenosis of Arteriovenous Access Through Wavelet Transform and Machine Learning on Sounds Recorded with an Electronic Stethoscope.	E	E	E	E
R0546	Deep learning-based frame synthesis enables radiation dose reduction in digital subtraction angiography imaging: a multicenter study	I	I	I	I
R0548	Auxiliary learning for anatomical localization of hierarchical vascular structures in digital subtraction angiography	I	I	I	I
R0550	Stent-RD-UNet: A Deep Learning Network for Real-Time Pose Estimation and Tracking of Aortic Stents in Image-Assisted Interventions	I	I	I	I
R0554	Automated Coronary Artery Disease Detection Using a CNN Model with Temporal Attention	I	I	I	I
R0559	Explainable vision transformer (E-VIT) based accurate diagnosis of stenosis in X-ray coronary angiography images (XCA)	E	E	E	E
R0569	Evaluation of the Diagnostic Utility of VNC Based AI-DSA Imaging for Follow up of Unruptured Intracranial Aneurysms	E	E	E	E
R0570	Development and application of AI enabled automated drug delivery system for TACE guided by DSA	I	I	I	I
R0574	Digital sales automation and buyer engagement: investigating the mediating role of perceived value in B2B exchanges	E	E	E	E
R0579	An AI-Driven Multimodal Framework for Interview Readiness Assessment and Skill Enhancement	E	E	E	E
R0581	Quantum-AI Number-Theoretic Frameworks for Post-Quantum Cryptography and Long-Term Information Security	E	E	E	E
R0584	From Prompts to Pipelines: Evaluating LLM-Generated Medical Image Segmentation Baselines	E	E	E	E
R0587	Anatomy-Informed Deep Learning for Abdominal Aortic Aneurysm Segmentation	E	E	E	E
R0598	Long-term effectiveness and safety of ablation index-guided high-power ablation combined with impedance spike cut-off in patients with atrial fibrillation.	E	E	E	E
R0599	Modelling C-arm fluoroscopy and operating table kinematics via machine learning.	E	E	E	E
R0602	The Role of Artificial Intelligence in Diagnosing Pulmonary Embolism: A Systematic Review and Meta-analysis.	E	E	E	E
R0612	New Insights into Molecular Mechanisms and Radiomics in Non-Contrast CT for Aortic Dissection: A Case Report and Literature Review.	E	E	E	E
R0618	CatLet Score as a Predictor of Cardiac Death in Patients With Acute Myocardial Infarction: Insights From Interpretable Machine Learning Models.	E	E	E	E
R0634	Knowledge-enhanced AI drives diagnosis of multiple retinal diseases in fundus fluorescein angiography.	E	E	E	E
R0635	Factors Impacting the Performance of Deep Learning Detection of Pulmonary Emboli.	E	E	E	E
R0640	Texture-Based Preprocessing Framework with nnU-Net Model for Accurate Intracranial Artery Segmentation.	I	I	I	I

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Table S1 – *continued from previous page*

Record	Title	R1	R2	Final	Pipeline
R0643	Multimodal machine learning for 5-year mortality prediction after percutaneous coronary intervention.	I	I	I	I
R0665	DDU-Net: learning complex vascular topologies with KAN-Swin transformers and double dynamic upsampler.	E	E	E	E
R0673	AI-Based Denoising and Identification of Neuromodulation Lead Locations from Limited Intraoperative Fluoroscopy Images.	E	E	E	E
R0676	Predicting Acute Ischemic Stroke Core in Multiphase CT Angiography Using a CT Perfusion-Trained Neural Network.	E	E	E	E
R0691	Application and Challenges of Deep Learning in Pulmonary Vessels Segmentation of CTPA Images.	E	E	E	E
R0721	Coronary Collaterals: A Comprehensive Review.	E	E	E	E
R0732	Automatic segmentation and reconstruction of lower-extremity arteries from computed tomography angiography images via a deep learning framework.	E	E	E	E
R0764	Leveraging Artificial Intelligence for the Diagnosis of Systemic Sclerosis Associated Pulmonary Arterial Hypertension: Opportunities, Challenges, and Future Perspectives.	E	E	E	E
R0774	End-to-end deep learning approach for uterine artery-ovarian artery anastomosis detection from digital subtraction angiography sequences.	I	I	I	I
R0777	Diagnosis model for assessing chronic thromboembolic pulmonary hypertension in high-altitude pulmonary embolism patients: a machine learning approach.	E	E	E	E
R0780	Automated AI detection of thoracic aortic dissection on CT imaging.	E	E	E	E
R0785	Extension of the GRACE score for non-ST-elevation acute coronary syndrome: a development and validation study in ten countries.	E	E	E	E
R0788	Artificial Intelligence-Assisted Low-Dose High Atomic Number Contrast Agent for Ultrahigh-Resolution Computed Tomography Angiography.	E	E	E	E
R0793	The Features of Vessel Densities and Hemorheological Parameters in Patients With Central Retinal Vein Occlusion: A Prospective Randomized Case-Control Study.	E	E	E	E
R0813	Impact of Deep Learning-based Artificial Intelligence Assistance on Reader Agreement in Coronary CT Angiography Interpretation.	E	E	E	E
R0814	Real-Time Artificial Intelligence-Assisted Middle Meningeal Artery Embolization Using Liquid Embolic Agents for Chronic Subdural Hematoma: A Preliminary Experience.	E	E	E	I
R0816	Clinical application of anterior segment optical coherence tomography in ocular emergencies: A comprehensive review.	E	E	E	E
R0824	Early auxiliary diagnosis model for chest pain triad based on artificial intelligence multimodal fusion.	E	E	E	E
R0828	Lesion detection in age-related macular degeneration with a multimodal imaging and machine learning approach.	E	E	E	E
R0841	Deep Learning-Based Diagnosis of Femoropopliteal Artery Steno-Occlusion Using Maximum Intensity Projection Images of CT Angiography.	E	E	E	E
R0849	Machine learning electrocardiography model to differentiate takotsubo syndrome from myocardial infarction.	E	E	E	E
R0859	Diagnostic Role of Radiology in Acute Gastrointestinal Bleeding: A Comprehensive Review.	E	E	E	E
R0861	Relations Between Pulsatility in the Optic Nerve Head or Peripapillary Retinal Vessels and the Rate of Progression in Glaucoma.	E	E	E	E
R0870	Metal artifact reduction from surgical clips for intracranial aneurysms in photon-counting detector CT angiography.	E	E	E	E
R0893	Machine Learning Prediction of Cardiovascular Risk in Type 1 Diabetes Mellitus Using Radiomic Features from Multimodal Retinal Images.	E	E	E	E
R0898	Multimodal dynamic hierarchical clustering model for post-stroke cognitive impairment prediction.	E	E	E	E
R0917	Using clinical and radiographic variables to predict intracranial aneurysm rupture status with machine learning.	E	E	E	E
R0920	Quality assessment of optical coherence tomography angiography images with Relative-distance-based Patch Distribution Modeling (R-PaDiM).	E	E	E	E

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Table S1 – *continued from previous page*

Record	Title	R1	R2	Final	Pipeline
R0934	Development and validation of machine learning models to predict vertebral artery injury by C2 pedicle screws.	E	E	E	E
R0935	Transformer-based self-supervised learning of pixel- and frequency-domain features for DMI grading on OCTA images.	E	E	E	E
R0938	Ablation Index-Guided Radiofrequency Ablation for Atrial Fibrillation: An Updated Meta-Analysis on Its Procedural Efficacy and Safety Profiles.	E	E	E	E
R0939	Gastrointestinal bleeding detection on digital subtraction angiography using convolutional neural networks with and without temporal information.	I	I	I	I
R0961	Application of non-invasive imaging in myocardial infarction: a bibliometric analysis from January 2003 to December 2022.	E	E	E	E
R0968	Failure of standard contemporary ST-elevation myocardial infarction electrocardiogram criteria to reliably identify acute occlusion of the left anterior descending coronary artery.	E	E	E	E
R1005	Assessment of cognitive load in the context of neurosurgery.	E	E	E	E
R1014	Ves-GAN: Unsupervised Vessel-Targeted Low-Dose Coronary Computed Tomography Angiography Denoising Framework.	E	E	E	E
R1045	Machine learning-based estimation of occupational radiation dose in interventional cardiology.	E	E	E	E
R1053	Coronary artery disease classification using ConvMixer based classifier from CT angiography images.	E	E	E	E
R1074	Automated segmentation of thoracic aortic lumen and vessel wall on three-dimensional bright- and black-blood magnetic resonance imaging using nnU-Net.	E	E	E	E
R1094	The Emerging Role of Computed Tomography Coronary Angiography in the Left Main Stem Percutaneous Coronary Intervention.	E	E	E	E
R1095	Research on ischemic stroke risk assessment based on CTA radiomics and machine learning.	E	E	E	E
R1106	Machine learning-based hemodynamics quantitative assessment of pulmonary circulation using computed tomographic pulmonary angiography.	E	E	E	E
R1108	Non-destructive detection of early wheat germination via deep learning-optimized terahertz imaging.	E	E	E	E
R1117	Real-Time Coronary Artery Dominance Classification from Angiographic Images Using Advanced Deep Video Architectures.	I	I	I	I
R1124	Training a deep learning model to predict the anatomy irradiated in fluoroscopic x-ray images.	E	E	E	E
R1157	Diagnostic Accuracy of On-Premise Automated Coronary CT Angiography Analysis Based on Coronary Artery Disease Reporting and Data System 2.0.	E	E	E	E
R1164	Novel artificial intelligence approach in neurointerventional practice: Preliminary findings on filter movement and ischemic lesions in carotid artery stenting.	E	E	E	I
R1168	Lipoproteins predicting coronary lesion complexity in premature coronary artery disease: a supervised machine learning approach.	E	E	E	E
R1173	Artificial intelligence software in biomedical imaging: a pharmaceutical perspective on radiology and contrast-enhanced ultrasound applications.	E	E	E	E
R1195	Application of deep learning reconstruction combined with time-resolved post-processing method to improve image quality in CTA derived from low-dose cerebral CT perfusion data.	E	E	E	E
R1209	perfDSA: Automatic Perfusion Imaging in Cerebral Digital Subtraction Angiography.	I	I	I	I
R1213	Multitask Deep Learning for Automated Detection of Endoleak at Digital Subtraction Angiography during Endovascular Aneurysm Repair.	I	I	I	I
R1222	LT-YOLO: long-term temporal enhanced YOLO for stenosis detection on invasive coronary angiography.	I	I	I	I
R1238	Optimizing Input Selection for Cardiac Model Training and Inference: An Efficient 3D Convolutional Neural Networks-Based Approach to Automate Coronary Angiogram Video Selection.	I	I	I	I
R1250	Implementation of a national AI technology program on cardiovascular outcomes and the health system.	E	E	E	E
R1251	Linking sequence restoration capability of shuffled coronary angiography to coronary artery disease diagnosis.	I	I	I	I

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Table S1 – *continued from previous page*

Record	Title	R1	R2	Final	Pipeline
R1263	Minimizing Human-Induced Variability in Quantitative Angiography for Robust and Explainable AI-Based Occlusion Prediction.	I	I	I	I
R1283	Robust resolution improvement of 3D UTE-MR angiogram of normal vasculatures using super-resolution convolutional neural network.	E	E	E	E
R1295	Intraoperative stenosis detection in X-ray coronary angiography via temporal fusion and attention-based CNN.	I	I	I	I
R1300	Deep learning based automatic quantification of aortic valve calcification on contrast enhanced coronary CT angiography.	E	E	E	E
R1305	Non-invasive derivation of instantaneous free-wave ratio from invasive coronary angiography using a new deep learning artificial intelligence model and comparison with human operators' performance.	E	E	E	E
R1317	Deep learning based coronary vessels segmentation in X-ray angiography using temporal information.	I	I	I	I
R1322	MIP-Enhanced Uncertainty-Aware Network for Fast 7T Time-of-Flight MRA Reconstruction.	E	E	E	E
R1325	Advancements in Computed Tomography Angiography for Pulmonary Embolism Assessment.	E	E	E	E
R1328	Beyond the eye: A relational model for early dementia detection using retinal OCTA images.	E	E	E	E
R1329	CVFSNet: A Cross View Fusion Scoring Network for end-to-end mTICI scoring.	I	I	I	I
R1341	Accurate and robust segmentation of cerebral distal small arteries by DVNet with dual contextual path and vascular attention enhancement.	E	E	E	E
R1344	Diagnostic value of deep learning reconstruction-based subtraction CT-FFR in patients with calcified-related stenosis or stent implantation.	E	E	E	E
R1357	Quantitative analysis of deep learning reconstruction in CT angiography: Enhancing CNR and reducing dose.	E	E	E	E
R1362	World's First Real-Time Artificial Intelligence-Assisted Mechanical Thrombectomy for Acute Ischemic Stroke.	I	I	I	I
R1367	Dual-stream disentangled model for microvascular extraction in five datasets from multiple OCTA instruments.	E	E	E	E
R1368	CDKD-w+: A Keyframe Recognition Method for Coronary Digital Subtraction Angiography Video Sequence Based on w+ Space Encoding.	I	I	I	I
R1374	Deep learning-assisted diagnosis of acute mesenteric ischemia based on CT angiography images.	E	E	E	E
R1376	Technique selection and technical developments for 2D dual-energy subtraction angiography on an interventional C-arm.	E	E	E	I
R1385	Multi-scale dual attention embedded U-shaped network for accurate segmentation of coronary vessels in digital subtraction angiography.	E	E	E	E
R1389	A continuous-action deep reinforcement learning-based agent for coronary artery centerline extraction in coronary CT angiography images.	E	E	E	E
R1399	Pericoronary adipose tissue feature analysis in computed tomography calcium score images in comparison to coronary computed tomography angiography.	E	E	E	E
R1421	Automated stenosis estimation of coronary angiographies using end-to-end learning.	I	I	I	I
R1422	Deep Learning Models for Automatic Classification of Anatomic Location in Abdominopelvic Digital Subtraction Angiography.	I	I	I	I
R1436	AutoFOX: An automated cross-modal 3D fusion framework of coronary X-ray angiography and OCT.	E	I	E†	I
R1441	The value of dual-energy computed tomography angiography-based virtual monoenergetic imaging for evaluations after cerebral aneurysm clipping.	E	E	E	E
R1503	Rupture risk assessment in cerebral arteriovenous malformations: an ensemble model using hemodynamic and morphological features.	I	E	I†	E
R1504	Large-scale pretrained frame generative model enables real-time low-dose DSA imaging: An AI system development and multicenter validation study.	I	I	I	I

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Table S1 – *continued from previous page*

Record	Title	R1	R2	Final	Pipeline
R1537	Enhanced Cerebrovascular Extraction Using Vessel-Specific Preprocessing of Time-Series Digital Subtraction Angiograph	I	I	I	I
R1541	OccluNet: Spatio-Temporal Deep Learning for Occlusion Detection on DSA	I	I	I	I
R1546	Machine Learning-Based Retinal Neovascularization Localization in En Face Optical Coherence Tomography Angiography Images Using Vessel Features	E	E	E	E
R1565	Reconstruction of 3D vascular flow patterns from sparse angiographic data using a 3D convolutional neural network (CNN)	I	I	I	I
R1566	Prediction of Rapidly Progressing Coronary Plaques Using a 3D Convolutional Neural Network Model Based on Coronary CT Angiography	E	E	E	E
R1599	HAResformer: A Hybrid ResNet-Transformer Hierarchical Aggregation Architecture for Visible-Infrared Person Reidentification	E	E	E	E
R1602	Automated Compliance Monitoring: A Machine Learning Approach for Digital Services Act Adherence in Multi-Product Platforms	E	E	E	E
R1603	High-resolution multiplexed antibody-omics and interpretable machine learning unveil novel pathogenic mechanisms in kidney transplant rejection	E	E	E	E
R1609	DSA-NRP: No-Reflow Prediction from Angiographic Perfusion Dynamics in Stroke EVT	I	I	I	I
R1614	Early prediction of student performance based on behavioral data in blended learning	E	E	E	E
R1620	Classifying Scam Calls Through Content Analysis With Dynamic Sparsity Top-k Attention Regularization	E	E	E	E
R1673	Coronary Artery Vessel Tree Segmentation Using Transfer Learning from CT Angiography Images	E	E	E	E
R1674	Multi-Frame Coronary Vessel Segmentation in X-Ray Angiography	I	I	I	I
R1675	Dynamic-Computed Tomography Angiography for Cerebral Vessel Templates and Segmentation	E	E	E	E
R1680	Integrated Pipeline for Coronary Angiography With Automated Lesion Profiling, Virtual Stenting, and 100-Vessel FFR Validation	I	I	I	I
R1694	Structure and oxygen saturation recovery of sparse photoacoustic microscopy images by deep learning	E	E	E	E
R1714	An Accurate Aneurysm Detection Model based on Artificial Intelligence	E	E	E	E
R1720	Deep Learning for Risky Cardiovascular and Cerebrovascular Event Prediction in Hypertensive Patients	E	E	E	E
R1725	FLOW LAYER HETEROGENEITY LEADING PLAQUE ECCENTRICITY IN CORONARY ARTERIES: DYNAMIC ANGIOGRAPHY WITH IVUS AND DEEP LEARNING ANALYSIS	I	I	I	I
R1740	Reducing motion artifacts in craniocervical background subtraction angiography with deformable registration and unsupervised deep learning.	I	I	I	I
R1765	The Role of Coronary Computed Tomography Angiography in the Diagnosis, Risk Stratification, and Management of Patients with Diabetes and Chest Pain.	E	E	E	E
R1774	Left, Then Right Internal Carotid Artery Dissection: A Case Report.	E	E	E	E
R1777	The protective role of gamma zone peripapillary atrophy in diabetic retinopathy: insights from deep learning and SS-OCT angiography.	E	E	E	E
R1784	The Role of Cardiac Computed Tomography Angiography in Risk Stratification for Coronary Artery Disease.	E	E	E	E
R1790	Impact of Leg Position on Measurements Used to Detect Femoral Component Subsidence in THA.	E	E	E	E
R1795	Generative modeling of the Circle of Willis using 3D-StyleGAN.	E	E	E	E
R1800	Deep learning model for automated diagnosis of moyamoya disease based on magnetic resonance angiography.	E	E	E	E
R1804	Diagnostic model of microvasculature and neurologic alterations in the retina and optic disc for lupus nephritis.	E	E	E	E
R1812	Artificial intelligence-based prediction of neurocardiovascular risk score from retinal swept-source optical coherence tomography-angiography.	E	E	E	E

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Table S1 – *continued from previous page*

Record	Title	R1	R2	Final	Pipeline
R1832	Deep-learning reconstruction enhances image quality of Adamkiewicz Artery in low-keV dual-energy CT.	E	E	E	E
R1833	First In-Vivo demonstration of 1000fps High Speed Coronary Angiography (HSCA) in a swine animal model.	E	E	E	I
R1841	The role of artificial intelligence in coronary CT angiography.	E	E	E	E
R1846	Utility of automated CT perfusion software in acute ischemic stroke with large and medium vessel occlusion.	E	E	E	E
R1849	Non-invasive physiological assessment of coronary artery obstruction on coronary computed tomography angiography.	E	E	E	E
R1854	Retinal vascular changes after Silicon Oil removal in the Eye with Rhegmatogenous Retinal detachment.	E	E	E	E
R1855	Exploring single-mode VCSEL wavelength tuning for low-cost 3D optical coherence tomography and OCT angiography.	E	E	E	E
R1862	Simulation-informed learning for time-resolved angiographic contrast agent concentration reconstruction.	I	I	I	I
R1879	Harnessing deep learning for detection of diabetic retinopathy in geriatric group using optical coherence tomography angiography-OCTA: A promising approach.	E	E	E	E
R1891	Focal cortical dysplasia lesion segmentation using multiscale transformer.	E	E	E	E
R1917	Dynamic change of acetabular component position in total hip arthroplasty based on the spinopelvic classification: a prospective radiographic study.	E	E	E	E
R1926	Integrated deep learning model for automatic detection and classification of stenosis in coronary angiography.	E	E	E	E
R1927	Validation of angiography-derived Murray law-based quantitative flow reserve (μQFR) against pressure-derived instantaneous wave-free ratio for assessing coronary lesions, a single-center study in Egypt.	E	E	E	E
R1936	Linking disease activity with optical coherence tomography angiography in neovascular age related macular degeneration using artificial intelligence.	E	E	E	E
R1939	Spatiotemporal Disentanglement of Arteriovenous Malformations in Digital Subtraction Angiography.	I	I	I	I
R1957	Automated identification of thrombectomy amenable vessel occlusion on computed tomography angiography using deep learning.	E	E	E	E
R1959	Perfusion parameter map generation from TOF-MRA in stroke using generative adversarial networks.	E	E	E	E
R1978	Noninvasive and fast method of calculation for instantaneous wave-free ratio based on haemodynamics and deep learning.	E	E	E	E
R1988	Optimizing neurointerventional procedures: an algorithm for embolization coil detection and automated collimation to enable dose reduction.	E	E	E	E
R2001	Artificial intelligence-based automatic nidus segmentation of cerebral arteriovenous malformation on time-of-flight magnetic resonance angiography.	E	E	E	E
R2007	Discovering hidden patterns: Association rules for cardiovascular diseases in type 2 diabetes mellitus.	E	E	E	E
R2013	Fully Automatic Quantitative Measurement of Equilibrium Radionuclide Angiocardiography Using a Convolutional Neural Network.	E	E	E	E
R2022	A Convolutional-Transformer Model for FFR and iFR Assessment From Coronary Angiography.	I	I	I	I
R2026	Automatic detection of pulmonary embolism on computed tomography pulmonary angiogram scan using a three-dimensional convolutional neural network.	E	E	E	E
R2043	Coronary computed tomographic angiography-derived anatomic and hemodynamic plaque characteristics in prediction of cardiovascular events.	E	E	E	E
R2054	Fluorescence and tracers in surgery: the coming future.	E	E	E	E
R2058	Cardiogenic shock trajectories: is the Society for Cardiovascular Angiography and Interventions definition the right one?	E	E	E	E
R2059	Development of a Novel Statistical Model for Predicting Clinical Outcomes in Stroke Patients With Tandem Occlusions After Endovascular Therapy.	E	E	E	E

Continued on next page

Table S1 – *continued from previous page*

Record	Title	R1	R2	Final	Pipeline
R2069	Super resolution deep learning reconstruction for coronary CT angiography: A structured phantom study.	E	E	E	E
R2073	Artificial intelligence in Medicare: utilization, spending, and access to AI-enabled clinical software.	E	E	E	E
R2083	Deep learning-based fully automatic screening of carotid artery plaques in computed tomography angiography: a multicenter study.	E	E	E	E
R2087	Evaluation of stenoses using AI video models applied to coronary angiography.	I	I	I	I
R2089	Real-time coronary artery segmentation in CAG images: A semi-supervised deep learning strategy.	E	E	E	I
R2091	Predicting major adverse cardiovascular events within 3 years by optimization of radiomics model derived from pericoronary adipose tissue on coronary computed tomography angiography: a case-control study.	E	E	E	E
R2101	Deep learning segmentation of non-perfusion area from color fundus images and AI-generated fluorescein angiography.	E	E	E	E
R2112	Automated Detection of Steno-Occlusive Lesion on Time-of-Flight MR Angiography: An Observer Performance Study.	E	E	E	E
R2114	CAVE: Cerebral artery-vein segmentation in digital subtraction angiography.	I	I	I	I
R2115	Illusion of revascularization: does anyone achieve optimal revascularization during percutaneous coronary intervention?	E	E	E	E
R2192	Emerging artificial intelligence-aided diagnosis and management methods for ischemic strokes and vascular occlusions: A comprehensive review.	E	E	E	E
R2201	Automatic identification of hypertension and assessment of its secondary effects using artificial intelligence: A systematic review (2013-2023).	E	E	E	E
R2208	Radiomics-Based Prediction of Collateral Status from CT Angiography of Patients Following a Large Vessel Occlusion Stroke.	E	E	E	E
R2213	Efficient management of pulmonary embolism diagnosis using a two-step interconnected machine learning model based on electronic health records data.	E	E	E	E
R2218	A novel hybrid machine learning model for auxiliary diagnosing myocardial ischemia.	E	E	E	E
R2222	Cardiac CT: Competition, complimentary or confounder.	E	E	E	E
R2225	Deep Learning Imaging Reconstruction Algorithm for Carotid Dual Energy CT Angiography: Opportunistic Evaluation of Cervical Intervertebral Discs-A Preliminary Study.	E	E	E	E
R2227	Cross-modality Labeling Enables Noninvasive Capillary Quantification as a Sensitive Biomarker for Assessing Cardiovascular Risk.	E	E	E	E
R2228	Automatic Detection of 30 Fundus Diseases Using Ultra-Widefield Fluorescein Angiography with Deep Experts Aggregation.	E	E	E	E
R2231	Effect of deep learning image reconstruction with high-definition standard scan mode on image quality of coronary stents and arteries.	E	E	E	E
R2241	OCT angiography 2023 update: focus on diabetic retinopathy.	E	E	E	E
R2243	A self-supervised learning approach for registration agnostic imaging models with 3D brain CTA.	E	E	E	E
R2249	Resilient back-propagation machine learning-based classification on fundus images for retinal microaneurysm detection.	E	E	E	E
R2254	Radiation reduction for interventional radiology imaging: a video frame interpolation solution.	I	I	I	I
R2263	Background Subtraction Angiography with Deep Learning Using Multi-frame Spatiotemporal Angiographic Input.	I	I	I	I
R2269	Enhancing the Prediction of Cardiac Allograft Vasculopathy Using Intravascular Ultrasound and Machine Learning: A Proof of Concept.	E	E	E	E
R2289	Interpretable Detection of Diabetic Retinopathy, Retinal Vein Occlusion, Age-Related Macular Degeneration, and Other Fundus Conditions.	E	E	E	E
R2311	Novel Techniques in Imaging Congenital Heart Disease: JACC Scientific Statement.	E	E	E	E

Continued on next page

Table S1 – *continued from previous page*

Record	Title	R1	R2	Final	Pipeline
R2324	Artificial intelligence and hemodynamic studies in optical coherence tomography angiography for diabetic retinopathy evaluation: A review.	E	E	E	E
R2334	The current and future role of imaging of thoracic aortic disease: a North American society for cardiovascular imaging commentary on the 2022 AHA/ACC guidelines for the diagnosis and management of aortic disease.	E	E	E	E
R2335	VET: Vasculature Extraction Transformer for Single-Scan Optical Coherence Tomography Angiography.	E	E	E	E
R2336	CASMatching strategy for automated detection and quantification of carotid artery stenosis based on digital subtraction angiography.	E	E	E	E
R2339	Improving the depiction of small intracranial vessels in head computed tomography angiography: a comparative analysis of deep learning reconstruction and hybrid iterative reconstruction.	E	E	E	E
R2366	Deep learning-based scan range optimization can reduce radiation exposure in coronary CT angiography.	E	E	E	E
R2376	Automated catheter segmentation and tip detection in cerebral angiography with topology-aware geometric deep learning.	I	I	I	I
R2377	Locating X-Ray Coronary Angiogram Keyframes via Long Short-Term Spatiotemporal Attention With Image-to-Patch Contrastive Learning.	I	I	I	I
R2394	Integrated deep learning approach for automatic coronary artery segmentation and classification on computed tomographic coronary angiography	E	E	E	E
R2396	Deep Learning for CAD Prediction: X-ray Angiography Insights	E	E	E	E
R2400	Unsupervised deep learning for the matching of vascular anatomy in multiple digital subtraction angiograms	I	I	I	I
R2411	Auxiliary Input in Training: Incorporating Catheter Features into Deep Learning Models for ECG-Free Dynamic Coronary Roadmapping	I	I	I	I
R2415	A Patient-Specific Registration of Coronary Angiogram-Fluoroscopy by Similarity- Based Transfer Learning	I	I	I	I
R2430	LB-3 | Artificial Intelligence for Extracting Non-COronary Data from Angiography: The AI-ENCODE Study	I	I	I	I
R2434	Optimizing Input Selection for Cardiac Model Training and Inference: An Efficient 3D CNN-based Approach to Automate Coronary Angiogram Video Selection	I	I	I	I
R2435	Detection of Intracranial Hemorrhage – A Comparative Study of Traditional Machine Learning Techniques and Mask Generation using CNN	E	E	E	E
R2443	FeDETR: A Federated Approach for Stenosis Detection in Coronary Angiography	I	I	I	I
R2466	4DRGS: 4D Radiative Gaussian Splatting for Efficient 3D Vessel Reconstruction from Sparse-View Dynamic DSA Images	I	I	I	I
R2468	Interventional guidewire segmentation algorithm based on coded deep neural networks	E	E	E	E
R2471	LinNet: Linear Network for Efficient Point Cloud Representation Learning	E	E	E	E
R2490	GaraMoSt: Parallel Multi-Granularity Motion and Structural Modeling for Efficient Multi-Frame Interpolation in DSA Images	I	I	I	I
R2491	MoSt-DSA: Modeling Motion and Structural Interactions for Direct Multi-Frame Interpolation in DSA Images	I	I	I	I
R2499	From PINNs to PIKANs: recent advances in physics-informed machine learning	E	E	E	E
R2513	Developing an artificial intelligence literacy framework: Evaluation of a literacy course for senior secondary students using a project-based learning approach	E	E	E	E
R2521	Artificial intelligence and edge computing for machine maintenance-review	E	E	E	E
R2526	Artificial intelligence in medicine: mitigating risks and maximizing benefits via quality assurance, quality control, and acceptance testing	E	E	E	E
R2547	Vascular Segmentation of Functional Ultrasound Images using Deep Learning	E	E	E	E
R2588	TCTAP A-037 Coronary Angiography Deep Learning Model for Diagnosis of Neoatheroslcerosis	E	E	E	E

Continued on next page

Table S1 – *continued from previous page*

Record	Title	R1	R2	Final	Pipeline
R2622	Identification and risk stratification of coronary disease by artificial intelligence-enabled ECG.	E	E	E	E
R2638	Machine Learning for Multi-Vessel Coronary Artery Disease Prediction on Electrocardiogram Gated Single-Photon Emission Computed Tomography.	E	E	E	E
R2642	OCT-angiography based artificial intelligence-inferred fluorescein angiography for leakage detection in retina [Invited].	E	E	E	E
R2646	A Lightweight Deep Learning Framework for Automatic MRI Data Sorting and Artifacts Detection.	E	E	E	E
R2653	Deep Learning Unveils Hidden Angiography in Noncontrast CT Scans.	E	E	E	E
R2659	Determination of lipid-rich plaques by artificial intelligence-enabled quantitative computed tomography using near-infrared spectroscopy as reference.	E	E	E	E
R2660	Recent advances in the direct anterior approach to total hip arthroplasty: a surgeon's perspective.	E	E	E	E
R2675	A framework of myocardial bridge detection with x-ray angiography sequence.	I	I	I	I
R2725	Using Neural Networks Algorithm in Ischemic Stroke Diagnosis: A Systematic Review.	E	E	E	E
R2726	The use of artificial intelligence in interventional cardiology.	E	E	E	E
R2776	Deep learning for artery-vein classification in optical coherence tomography angiography.	E	E	E	E
R2786	Use of high-speed angiography HSA-derived boundary conditions and Physics Informed Neural Networks (PINNs) for comprehensive estimation of neurovascular hemodynamics.	I	I	I	I
R2803	Association of retinal microvascular abnormalities and neuromyelitis optica spectrum disorders with optical coherence tomography angiography.	E	E	E	E
R2828	Identification of ischemia-causing lesions using coronary plaque quantification and changes in fractional flow reserve derived from computed tomography across the lesion.	E	E	E	E
R2840	Coronary X-ray angiography segmentation using Artificial Intelligence: a multicentric validation study of a deep learning model.	E	E	E	E
R2845	Application of Pseudo-Three-Dimensional Residual Network to Classify the Stages of Moyamoya Disease.	I	I	I	I
R2851	ADSeg: A flap-attention-based deep learning approach for aortic dissection segmentation.	E	E	E	E
R2855	FSTIF-UNet: A Deep Learning-Based Method Towards Automatic Segmentation of Intracranial Aneurysms in Un-Reconstructed 3D-RA.	I	I	I	I
R2859	Accuracy of artificial intelligence software for CT angiography in stroke.	E	E	E	E
R2860	Radiomics Signatures of Carotid Plaque on Computed Tomography Angiography : An Approach to Identify Symptomatic Plaques.	E	E	E	E
R2873	Automated Assessment of Cardiac Systolic Function From Coronary Angiograms With Video-Based Artificial Intelligence Algorithms.	I	I	I	I
R2882	Artificial intelligence for advanced analysis of coronary plaque.	E	E	E	E
R2891	Retinal imaging technologies in cerebral malaria: a systematic review.	E	E	E	E
R2933	An All-in-One Tool for 2D Atherosclerotic Disease Assessment and 3D Coronary Artery Reconstruction.	E	E	E	E
R2976	Deep learning image reconstruction algorithm: impact on image quality in coronary computed tomography angiography.	E	E	E	E
R2977	DEEP MOVEMENT: Deep learning of movie files for management of endovascular thrombectomy.	I	I	I	I
R2985	Auto-Detection of Motion Artifacts on CT Pulmonary Angiograms with a Physician-Trained AI Algorithm.	E	E	E	E
R2992	Deep learning-based end-to-end automated stenosis classification and localization on catheter coronary angiography.	I	I	I	I
R3016	Improving the diagnostic performance of computed tomography angiography for intracranial large arterial stenosis by a novel super-resolution algorithm based on multi-scale residual denoising generative adversarial network.	E	E	E	E

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Table S1 – *continued from previous page*

Record	Title	R1	R2	Final	Pipeline
R3030	Coronary artery stenosis detection via proposal-shifted spatial-temporal transformer in X-ray angiography.	I	I	I	I
R3039	Deep learning-based prediction of future myocardial infarction using invasive coronary angiography: a feasibility study.	E	E	E	E
R3042	A token-mixer architecture for CAD-RADS classification of coronary stenosis on multiplanar reconstruction CT images.	E	E	E	E
R3050	Toward bedside computation of myocardial infarction risk using noninvasive analysis of patients living with coronary artery disease.	E	E	E	E
R3052	Radial Wall Strain Assessment From AI-Assisted Angiography: Feasibility and Agreement With OCT as Reference Standard.	I	I	I	I
R3063	Jigsaw training-based background reverse attention transformer network for guidewire segmentation.	I	I	I	I
R3067	Mapping of dust source susceptibility by remote sensing and machine learning techniques (case study: Iran-Iraq border).	E	E	E	E
R3093	Automatic assessment of collaterals physiology in chronic total occlusions by means of artificial intelligence.	I	I	I	I
R3107	Correlation of vascular and fluid-related parameters in neovascular age-related macular degeneration using deep learning.	E	E	E	E
R3112	Segmentation of macular neovascularization and leakage in fluorescein angiography images in neovascular age-related macular degeneration using deep learning.	E	E	E	E
R3126	Angular super-resolution in X-ray projection radiography using deep neural network: Implementation on rotational angiography.	I	I	I	I
R3128	Angiography-based coronary flow reserve: The feasibility of automatic computation by artificial intelligence.	I	I	I	I
R3129	Learning spatiotemporal features of DSA using 3D CNN and Bi-ConvGRU for ischemic moyamoya disease detection.	I	I	I	I
R3142	ConvNeXtv2 Fusion with Mask R-CNN for Automatic Region Based Coronary Artery Stenosis Detection for Disease Diagnosis	E	E	E	E
R3146	Structurally Different Neural Network Blocks for the Segmentation of Atrial and Aortic Perivascular Adipose Tissue in Multi-centre CT Angiography Scans	E	E	E	E
R3149	ConTrack: Contextual Transformer for Device Tracking in X-ray	I	I	I	I
R3155	Towards clinical translation of deep-learning based classification of DSA image sequences for stroke treatment	I	I	I	I
R3164	TiAVox: Time-aware Attenuation Voxels for Sparse-view 4D DSA Reconstruction	I	I	I	I
R3186	A neural network-enhanced methodology for the rapid establishment of local DRLs in interventional radiology: EVAR as a case example.	E	E	E	E
R3190	Abstract 17051: Oxytocin Receptor in the Spinal Neuraxial Modulation of Ventricular Excitability	E	E	E	E
R3194	NCIRF: an organ dose calculator for patients undergoing radiography and fluoroscopy	E	E	E	E
R3198	Multicenter aortic vessel tree extraction using deep learning	E	E	E	E
R3199	Abstract TP88: Deep Learning-based Automatic Detection And Segmentation Of Hyperdense Artery Sign On Non-contrast Computed Tomography In Acute Ischemic Stroke	E	E	E	E
R3212	Attention adaptive instance normalization style transfer for vascular segmentation using deep learning	E	E	E	E
R3234	Prolonged turbulence and deposition of LDL cholesterol caused atherosclerotic lesion in femoral arteries and spared the iliac arteries. Why? A dynamic angiography and machine learning analysis	E	E	E	I
R3239	Global relationship memory network for retinal capillary segmentation on optical coherence tomography angiography images	E	E	E	E
R3285	Real-time 3D echocardiographic transilluminated imaging combined with artificially intelligent left atrial appendage measurement for atrial fibrillation interventional procedures.	E	E	E	E
R3301	Machine Learning for Aiding Blood Flow Velocity Estimation Based on Angiography.	E	E	E	I
R3329	Machine Learning From Quantitative Coronary Computed Tomography Angiography Predicts Fractional Flow Reserve-Defined Ischemia and Impaired Myocardial Blood Flow.	E	E	E	E

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Table S1 – *continued from previous page*

Record	Title	R1	R2	Final	Pipeline
R3330	Computed Tomography Angiography and B-Mode Ultrasonography under Artificial Intelligence Plaque Segmentation Algorithm in the Perforator Localization for Preparation of Free Anterolateral Femoral Flap.	E	E	E	E
R3338	Self-supervised learning enables 3D digital subtraction angiography reconstruction from ultra-sparse 2D projection views: A multicenter study.	I	I	I	I
R3357	Detecting Coronary Artery Disease from Computed Tomography Images Using a Deep Learning Technique.	E	E	E	E
R3360	X-ray coronary centerline extraction based on C-UNet and a multifactor reconnection algorithm.	E	E	E	E
R3370	Radial wall strain: a novel angiographic measure of plaque composition and vulnerability.	E	E	E	I
R3375	Automatic landmark detection and mapping for 2D/3D registration with BoneNet.	E	E	E	E
R3376	Dataset on acute stroke risk stratification from CT angiographic radiomics.	E	E	E	E
R3388	Denoising using deep-learning-based reconstruction for whole-heart coronary MRA with sub-millimeter isotropic resolution at 3 T: a volunteer study.	E	E	E	E
R3391	Initial investigation of the use of angiographic parametric imaging for early prognosis of delayed cerebral ischemia in patients with subarachnoid hemorrhage.	I	I	I	I
R3392	Quantitative angiography prognosis of intracranial aneurysm treatment failure using parametric imaging and distal vessel analysis.	I	I	I	I
R3399	Deep learning for quality assessment of optical coherence tomography angiography images.	E	E	E	E
R3421	Which one of LDL-C /HDL-C ratio and non-HDL-C can better predict the severity of coronary artery disease in STEMI patients.	E	E	E	E
R3426	eICAB: A novel deep learning pipeline for Circle of Willis multiclass segmentation and analysis.	E	E	E	E
R3454	Optical flow estimation of coronary angiography sequences based on semi-supervised learning.	I	I	I	I
R3460	CT in Transcatheter-delivered Treatment of Valvular Heart Disease.	E	E	E	E
R3469	Deep learning-based classification of DSA image sequences of patients with acute ischemic stroke.	I	I	I	I
R3485	Real-time automatic prediction of treatment response to transcatheter arterial chemoembolization in patients with hepatocellular carcinoma using deep learning based on digital subtraction angiography videos.	I	I	I	I
R3529	The auto segmentation for cardiac structures using a dual-input deep learning network based on vision saliency and transformer.	E	E	E	E
R3531	Impact of machine-learning-based coronary computed tomography angiography-derived fractional flow reserve on decision-making in patients with severe aortic stenosis undergoing transcatheter aortic valve replacement.	E	E	E	E
R3551	Rethinking the neighborhood information for deep learning-based optical coherence tomography angiography.	E	E	E	E
R3553	Using Text Content From Coronary Catheterization Reports to Predict 5-Year Mortality Among Patients Undergoing Coronary Angiography: A Deep Learning Approach.	E	E	E	E
R3555	Using a Convolutional Neural Network and Convolutional Long Short-term Memory to Automatically Detect Aneurysms on 2D Digital Subtraction Angiography Images: Framework Development and Validation.	I	I	I	I
R3558	Self-calibration of C-arm imaging system using interventional instruments during an intracranial biplane angiography.	E	E	E	E
R3561	Combined Coronary CT-Angiography and TAVI Planning: Utility of CT-FFR in Patients with Morphologically Ruled-Out Obstructive Coronary Artery Disease.	E	E	E	E
R3562	A machine learning-based clinical decision support algorithm for reducing unnecessary coronary angiograms.	E	E	E	E
R3565	Radiomics versus Conventional Assessment to Identify Symptomatic Participants at Carotid Computed Tomography Angiography.	E	E	E	E

Continued on next page

Table S1 – *continued from previous page*

Record	Title	R1	R2	Final	Pipeline
R3576	Automatic Identification of Bioprostheses on X-ray Angiographic Sequences of Transcatheter Aortic Valve Implantation Procedures Using Deep Learning.	I	I	I	I
R3600	Improved Stroke Care in a Primary Stroke Centre Using AI-Decision Support.	E	E	E	E
R3601	Spatio-temporal deep learning for automatic detection of intracranial vessel perforation in digital subtraction angiography during endovascular thrombectomy.	I	I	I	I
R3607	In Vitro Study of the Precision and Accuracy of Measurement of the Vascular Inner Diameter on Computed Tomography Angiography Using Deep Learning Image Reconstruction: Comparison With Filtered Back Projection and Iterative Reconstruction.	E	E	E	E
R3615	Automatic construction of coronary artery tree structure based on vessel blood flow tracking.	I	I	I	I
R3630	Influence of diabetes mellitus on the diagnostic performance of machine learning-based coronary CT angiography-derived fractional flow reserve: a multicenter study.	E	E	E	E
R3648	Deep Learning-Based Approach to Automatically Assess Coronary Distensibility Following Kawasaki Disease.	I	I	I	I
R3661	Identification of successful cerebral reperfusions (mTICI $\geq$2b) using an artificial intelligence strategy.	I	I	I	I
R3662	Optical coherence tomography (OCT) angiolytics: a review of OCT angiography quantitative biomarkers.	E	E	E	E
R3689	Noninvasive Machine Learning Screening Model for Dacryocystitis Based on Ocular Surface Indicators.	E	E	E	E
R3690	Diagnostic safety of a machine learning-based automatic patient selection algorithm for stress-only myocardial perfusion SPECT.	E	E	E	E
R3707	Spatio-Temporal U-Net for Cerebral Artery and Vein Segmentation in Digital Subtraction Angiography	I	I	I	I
R3710	Extracting heterogeneous vessels in X-ray coronary angiography via machine learning	E	E	E	I
R3712	Robust Landmark-based Stent Tracking in X-ray Fluoroscopy	I	I	I	I
R3723	[Development of the DSA Method for Coronary Angiography Using Deep Learning Techniques].	E	E	E	E
R3734	Research on A Chiplet-based DSA (Domain-Specific Architectures) Scalable Convolutional Acceleration Architecture	E	E	E	E
R3779	Algorithm for Retinal Vessel Segmentation based on Axial Transformer and Convolutional Neural Network	E	E	E	E
R3783	Cortical Vessel Segmentation for Neuronavigation Using Vesselness-Enforced Deep Neural Networks	E	E	E	E
R3802	A Risk-Stratification Machine Learning Framework for the Prediction of Coronary Artery Disease Severity: Insights From the GESS Trial.	E	E	E	E
R3818	Automated Segmentation of Trigeminal Nerve and Cerebrovasculature in MR-Angiography Images by Deep Learning.	E	E	E	E
R3820	Myocardial Perfusion Imaging After Severe COVID-19 Infection Demonstrates Regional Ischemia Rather Than Global Blood Flow Reduction.	E	E	E	E
R3831	Improved Centerline Extraction in Fully Automated Coronary Ostium Localization and Centerline Extraction Framework using Deep Learning.	E	E	E	E
R3849	Relationship of age, atherosclerosis and angiographic stenosis using artificial intelligence.	E	E	E	E
R3858	Autonomous lumbar spine pedicle screw planning using machine learning: A validation study.	E	E	E	E
R3862	Early microvascular changes in patients with prediabetes evaluated by optical coherence tomography angiography.	E	E	E	E
R3867	Generative adversarial networks improve interior computed tomography angiography reconstruction.	E	E	E	E
R3890	Robotics, imaging, and artificial intelligence in the catheterisation laboratory.	E	E	E	E
R3893	Application of Artificial Intelligence in Acute Coronary Syndrome: A Brief Literature Review.	E	E	E	E
R3900	Deep Learning-Based Automated Thrombolysis in Cerebral Infarction Scoring: A Timely Proof-of-Principle Study.	I	I	I	I

Continued on next page

Table S1 – *continued from previous page*

Record	Title	R1	R2	Final	Pipeline
R3910	Real-time respiratory motion compensated roadmaps for hepatic arterial interventions.	I	I	I	I
R3929	The Aneurysm Occlusion Assistant, an AI platform for real time surgical guidance of intracranial aneurysms.	E	I	E†	I
R3932	Integrative analysis of the choroid by quantifying Haller vessel and choriocapillaris parameters in different drusen subtypes.	E	E	E	E
R3933	Automated three-dimensional vessel reconstruction based on deep segmentation and bi-plane angiographic projections.	E	E	E	I
R3942	X-Ray cardiac angiographic vessel segmentation based on pixel classification using machine learning and region growing.	E	E	E	E
R3970	Temporal-Spatial Feature Extraction of DSA Video and Its Application in AVM Diagnosis.	I	I	I	I
R3981	Changes in Macular Microvascular Structure in Macular Edema Secondary to Branch Retinal Vein Occlusion Treated with Anti-vascular Endothelial Growth Factor for One Year.	E	E	E	E
R3983	Deep Learning Improves the Temporal Reproducibility of Aortic Measurement.	E	E	E	E
R3999	autoTICI: Automatic Brain Tissue Reperfusion Scoring on 2D DSA Images of Acute Ischemic Stroke Patients.	I	I	I	I
R4003	Semi-automatic vessel detection for challenging cases of peripheral arterial disease.	E	E	E	E
R4015	Automatic detection of leakage point in central serous chorioretinopathy of fundus fluorescein angiography based on time sequence deep learning.	E	E	E	E
R4029	Deep Learning-based Angiogram Generation Model for Cerebral Angiography without Misregistration Artifacts.	I	I	I	I
R4054	A 2-year investigation of the impact of the computed tomography-derived fractional flow reserve calculated using a deep learning algorithm on routine decision-making for coronary artery disease management.	E	E	E	E
R4087	Automated segmentation of retinal nonperfusion area in fluorescein angiography in retinal vein occlusion using convolutional neural networks.	E	E	E	E
R4089	ROSE: A Retinal OCT-Angiography Vessel Segmentation Dataset and New Model.	E	E	E	E
R4092	Convolutional Network With Twofold Feature Augmentation for Diabetic Retinopathy Recognition From Multi-Modal Images.	E	E	E	E
R4097	A Three-Dimensional Deep Convolutional Neural Network for Automatic Segmentation and Diameter Measurement of Type B Aortic Dissection.	E	E	E	E
R4102	Automatic stenosis recognition from coronary angiography using convolutional neural networks.	I	I	I	I
R4114	Development of Novel Artificial Intelligence to Detect the Presence of Clinically Meaningful Coronary Atherosclerotic Stenosis in Major Branch from Coronary Angiography Video.	I	I	I	I
R4115	Machine learning-aided risk stratification system for the prediction of coronary artery disease.	E	E	E	E
R4143	Deep-Learning-Based Registration of Diagnostic Angiogram and Live Fluoroscopy for Percutaneous Coronary Intervention	I	I	I	I
R4148	Stent enhancement using marker response network and focus conversion learning in x-ray fluoroscopy sequences	I	I	I	I
R4154	Using a Convolutional Neural Network and Convolutional Long Short-term Memory to Automatically Detect Aneurysms on 2D-DSA Images (Preprint)	I	I	I	I
R4158	Radio Propagation Estimation in a Long-Range Environment using a Deep Neural Network	E	E	E	E
R4204	Automated Deep Learning Analysis of Angiography Video Sequences for Coronary Artery Disease	I	I	I	I
R4206	Deep Learning-Based Photoacoustic Imaging of Vascular Network Through Thick Porous Media	E	E	E	E
R4208	Automatic vessel segmentation in X-ray angiogram using spatio-temporal fully-convolutional neural network	I	I	I	I
R4218	Multi-scale residual denoising GAN model for producing super-resolution CTA images	E	E	E	E
R4220	Analysis of Coronary Angiography Video Interpolation Methods to Reduce X-ray Exposure Frequency Based on Deep Learning	I	I	I	I

Continued on next page

Table S1 – *continued from previous page*

Record	Title	R1	R2	Final	Pipeline
R4238	High Response Rate and Prolonged Survival of Unresectable Biliary Tract Cancer Treated With a New Combination Therapy Consisting of Intraarterial Chemotherapy Plus Radiotherapy.	E	E	E	E
R4258	Mesopic and Scotopic Light Sensitivity and Its Microstructural Correlates in Pseudoxanthoma Elasticum.	E	E	E	E
R4259	Detection of subjects with ischemic heart disease by using machine learning technique based on heart rate total variability parameters.	E	E	E	E
R4263	Diabetic Retinopathy (DR) Severity Level Classification Using Multimodel Convolutional Neural Networks.	E	E	E	E
R4265	End-to-End Deep Learning Model for Cardiac Cycle Synchronization from Multi-View Angiographic Sequences.	I	I	I	I
R4278	The sub-millisievert era in CTCA: the technical basis of the new radiation dose approach.	E	E	E	E
R4280	Comparison of Support Vector Machine, Naïve Bayes and Logistic Regression for Assessing the Necessity for Coronary Angiography.	E	E	E	E
R4294	Isolated low-grade solid organ injuries in children following blunt abdominal trauma: Is it time to consider discharge from the emergency department?	E	E	E	E
R4317	Deep neural networks for ECG-free cardiac phase and end-diastolic frame detection on coronary angiographies.	I	I	I	I
R4323	Motion-flow-guided recurrent network for respiratory signal estimation of x-ray angiographic image sequences.	I	I	I	I
R4333	Automated coronary artery atherosclerosis detection and weakly supervised localization on coronary CT angiography with a deep 3-dimensional convolutional neural network.	E	E	E	E
R4339	Sequential vessel segmentation via deep channel attention network.	I	I	I	I
R4362	Usefulness of Denoising Process to Depict Myopic Choroidal Neovascularisation Using a Single Optical Coherence Tomography Angiography Image.	E	E	E	E
R4405	Quantitative lobar pulmonary perfusion assessment on dual-energy CT pulmonary angiography: applications in pulmonary embolism.	E	E	E	E
R4411	Dynamic coronary roadmapping via catheter tip tracking in X-ray fluoroscopy with deep learning based Bayesian filtering.	I	I	I	I
R4441	Automatic Diagnosis Based on Spatial Information Fusion Feature for Intracranial Aneurysm.	I	I	I	I
R4447	Identification and Quantification of Cardiovascular Structures From CCTA: An End-to-End, Rapid, Pixel-Wise, Deep-Learning Method.	E	E	E	E
R4458	Effect of surgical intraocular pressure lowering on retinal structures - nerve fibre layer, foveal avascular zone, peripapillary and macular vessel density: 1 year results.	E	E	E	E
R4459	iDHS-DSAMS: Identifying DNase I hypersensitive sites based on the dinucleotide property matrix and ensemble bagged tree.	E	E	E	E
R4468	Learning Depth from a Single 2D Digital Subtraction Angiography Image as Semantic Segmentation	I	I	I	I
R4470	Deep Learning Assisted Automatic Intra-operative 3D Aortic Deformation Reconstruction	E	E	E	E
R4475	Deep learning for cerebral DSA to free from misregistration artifacts.	I	I	I	I
R4477	Clinically Applicable Deep Learning for Intracranial Aneurysm Detection in Computed Tomography Angiography Images: A Comprehensive Multicohort Study	E	E	E	E
R4479	Fully automated intracranial aneurysm detection and segmentation from digital subtraction angiography series using an end-to-end spatiotemporal deep neural network	I	I	I	I
R4482	Optimization of DSA image data input to a machine learning aneurysm identifier	I	I	I	I
R4492	Parameterized Explainer for Graph Neural Network	E	E	E	E
R4504	Transcatheter CT Hepatic Arteriography Compared with Conventional CT Fluoroscopy Guidance in Percutaneous Thermal Ablation to Treat Colorectal Liver Metastases: A Single-Center Comparative Analysis of 2 Historical Cohorts.	E	E	E	E
R4521	Hybrid deep learning network for vascular segmentation in photoacoustic imaging.	E	E	E	E
R4522	Toward accurate quantitative photoacoustic imaging: learning vascular blood oxygen saturation in three dimensions	E	E	E	E

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Table S1 – *continued from previous page*

Record	Title	R1	R2	Final	Pipeline
R4527	AIFNet: Automatic Vascular Function Estimation for Perfusion Analysis Using Deep Learning	E	E	E	E
R4533	A topological encoding convolutional neural network for segmentation of 3D multiphoton images of brain vasculature using persistent homology	E	E	E	E
R4544	Analysis of Video Retinal Angiography With Deep Learning and Eulerian Magnification	E	E	E	E
R4546	Real-time retinal layer segmentation of adaptive optics optical coherence tomography angiography with deep learning	E	E	E	E
R4547	Deep learning convolutional networks for image quality assessment in ultra-widefield fluorescein angiography	E	E	E	E
R4557	Automatic detection on intracranial aneurysm from digital subtraction angiography with cascade convolutional neural networks.	I	E	I†	E
R4570	Deep vessel segmentation by learning graphical connectivity.	E	E	E	E
R4579	Automatic segmentation of arterial tree from 3D computed tomographic pulmonary angiography (CTPA) scans.	E	E	E	E
R4593	Supervised Machine Learning Based Multi-Task Artificial Intelligence Classification of Retinopathies.	E	E	E	E
R4596	On-Site Computed Tomography-Derived Fractional Flow Reserve Using a Machine-Learning Algorithm - Clinical Effectiveness in a Retrospective Multicenter Cohort.	E	E	E	E
R4598	A deep learning based pipeline for optical coherence tomography angiography.	E	E	E	E
R4632	Machine learning algorithms for predicting scapular kinematics.	E	E	E	E
R4637	Abdominal Aortic Aneurysm Segmentation Using Convolutional Neural Networks Trained with Images Generated with a Synthetic Shape Model.	E	E	E	E
R4648	Cycle-consistent adversarial denoising network for multiphase coronary CT angiography.	E	E	E	E
R4651	Artificial intelligence machine learning-based coronary CT fractional flow reserve (CT-FFR_{ML}): Impact of iterative and filtered back projection reconstruction techniques.	E	E	E	E
R4661	EUS-guided antegrade intervention for benign biliary diseases in patients with surgically altered anatomy (with videos).	E	E	E	E
R4666	Catheter Synthesis in X-Ray Fluoroscopy with Generative Adversarial Networks	E	E	E	E
R4671	Fully automated unruptured intracranial aneurysm detection and segmentation from digital subtraction angiography series using an end-to-end spatiotemporal deep neural network	I	I	I	I
R4673	Use of a convolutional neural network for aneurysm identification in digital subtraction angiography	E	E	E	E
R4682	Machbarkeitsstudie zur CNN-basierten Identifikation und TICI-Klassifizierung zerebraler ischämischer Infarkte in DSA-Daten	E	E	E	E
R4683	Feasibility study of deep neural networks to classify intracranial aneurysms using angiographic parametric imaging	I	I	I	I
R4693	Automated Machine Learning - Methods, Systems, Challenges	E	E	E	E
R4699	Fully Automatic Dual-Guidewire Segmentation for Coronary Bifurcation Lesion	I	I	I	I
R4726	Fully Convolved Neural Network-Based Retinal Vessel Segmentation with Entropy Loss Function	E	E	E	E
R4729	Retinal vascular segmentation combined with PCNN and morphological matching enhancement	E	E	E	E
R4730	Digital resolution enhancement in low transverse sampling optical coherence tomography angiography using deep learning	E	E	E	E
R4744	Automated ASPECT rating: comparison between the Frontier ASPECT Score software and the Brainomix software.	E	E	E	E
R4762	Real-Time Medical Video Denoising with Deep Learning: Application to Angiography.	I	I	I	I
R4763	A clinical and proteomics approach to predict the presence of obstructive peripheral arterial disease: From the Catheter Sampled Blood Archive in Cardiovascular Diseases (CASABLANCA) Study.	E	E	E	E
R4779	Prediction of rupture risk in anterior communicating artery aneurysms with a feed-forward artificial neural network.	E	E	E	E
R4780	Multimodal Imaging in Diabetic Macular Edema.	E	E	E	E
R4799	Evaluating Deep Neural Networks to Classify Modulated and Coded Radio Signals	E	E	E	E

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Table S1 – *continued from previous page*

Record	Title	R1	R2	Final	Pipeline
R4804	A Motion Artifact Reduction Method in Cerebrovascular DSA Sequence Images	I	I	I	I
R4811	Cone-Beam CT With Augmented Fluoroscopy Combined With Electromagnetic Navigation Bronchoscopy for Biopsy of Pulmonary Nodules	E	E	E	E
R4854	Predicting climate conditions using Internet-of-Things and artificial hydrocarbon networks	E	E	E	E
R4863	Accuracy of robot-guided versus freehand fluoroscopy-assisted pedicle screw insertion in thoracolumbar spinal surgery.	E	E	E	E
R4876	A Case Study: Analyzing City Vitality with Four Pillars of Activity-Live, Work, Shop, and Play.	E	E	E	E
R4878	Plasma proteomic analysis of stable coronary artery disease indicates impairment of reverse cholesterol pathway.	E	E	E	E
R4896	Decreased coronary blood flow velocity in patients with aortic insufficiency but normal coronary arteries: the use of TIMI frame count in aortic insufficiency cases.	E	E	E	I
R4915	Implementation of a Power Quality signal classification system using wavelet based energy distribution and neural network	E	E	E	E
R4916	A Neural Network based power quality signal classification system using wavelet energy distribution	E	E	E	E
R4920	Prediction of acute antibody mediated rejection in antibody incompatible renal transplantation using machine learning for wide data	E	E	E	E
R4923	OC-0082: A machine learning method demonstrates that a large number of SNPs contribute to clinical radiosensitivity	E	E	E	E
R4954	Objected constrained registration and manifold learning: a new patient setup approach in image guided radiation therapy of thoracic cancer.	E	E	E	E
R4955	Sellar region related to the transsphenoidal endoscopic approach of pituitary surgery.	E	E	E	E
R4962	Tumor tracking based on correlation models in scanned ion beam therapy: an experimental study	E	E	E	E
R4972	Model-based fusion of CT and non-contrasted 3D C-arm CT: Application to transcatheter valve therapies	E	E	E	E
R5005	Low-density lipoprotein (LDL), which includes apolipoprotein A-I (apoAI-LDL) as a novel marker of coronary artery disease.	E	E	E	E
R5009	Clinical-sonographic index (CSI): a novel transcranial Doppler diagnostic model for middle cerebral artery stenosis.	E	E	E	E
R5016	Arterial stiffness and Chlamydia pneumoniae infection in coronary artery disease. Is there a link?	E	E	E	E
R5020	Arterial wave reflection is associated with severity of extracoronary atherosclerosis in patients with coronary artery disease.	E	E	E	E
R5041	Lipoprotein lipase gene polymorphisms in Croatian patients with coronary artery disease.	E	E	E	E

Table S2: PRISMA 2020 checklist. Locations refer to section, figure, table and appendix numbers of the main manuscript.

Topic	Item	Checklist item	Location where reported
TITLE			
Title	1	Identify the report as a systematic review	Title
ABSTRACT			
Abstract	2	Structured summary (PRISMA 2020 for Abstracts)	Abstract
INTRODUCTION			
Rationale	3	Rationale in the context of existing knowledge	Sections 1, 1.1, 1.2
Objectives	4	Explicit statement of objectives	Section 1.3
METHODS			
Eligibility criteria	5	Inclusion and exclusion criteria and grouping for syntheses	Section 2.2; Table 1
Information sources	6	Databases and other sources, with date last searched	Section 2.1; Appendix A
Search strategy	7	Full search strategies, including filters and limits	Section 2.1; Appendix A
Selection process	8	Methods to decide eligibility, reviewers, independence, automation tools	Sections 2.3, 2.3.1, 2.3.2; Appendix B; Table S1
Data collection process	9	Methods to collect data, reviewers, independence, automation tools	Section 2.5; Appendix D
Data items	10a	Outcomes sought	Section 2.5; Appendix D
	10b	Other variables sought	Section 2.5; Appendix D
Study risk of bias assessment	11	Methods to assess risk of bias in included studies	Not assessed; reporting completeness against the CLAIM 2024 checklist assessed (Sections 2.5, 3.7; Appendix E); rationale given in Section 4.3
Effect measures	12	Effect measures used	Section 2.6 (descriptive summaries of reported metrics; no pooled effect measure)
Synthesis methods	13a	Deciding which studies were eligible for each synthesis	Section 2.6 (all 115 studies)
	13b	Data preparation for presentation or synthesis	Sections 2.5, 2.6; Appendix D
	13c	Tabulation and visual display of results	Section 2.6
	13d	Methods to synthesize results	Section 2.6 (descriptive; no meta-analysis); Section 4.3
	13e	Methods to explore heterogeneity	Section 2.6; Section 3.4.2 (cross-tabulation by task, modality, territory, year)
	13f	Sensitivity analyses	Not performed
Reporting bias assessment	14	Methods to assess risk of bias due to missing results	Not performed; possible publication bias discussed in Section 3.6
Certainty assessment	15	Methods to assess certainty in the body of evidence	Not performed (no pooled outcome)
RESULTS			
Study selection	16a	Search and selection results, ideally with a flow diagram	Section 3.1; Figure 1
	16b	Studies that seemed to meet criteria but were excluded, and why	Section 3.1; Table S1 lists the 14 LLM-eligible records rejected at dual review with both reviewers' decisions; per-record exclusion reasons were not recorded. At full text, 2 meeting abstracts were excluded (E1) and 3 earlier versions of included articles were merged (Section 3.1)
Study characteristics	17	Cite each included study and present its characteristics	Sections 3.2–3.7; included-study manifest and extraction records in the public repository (Data availability statement)
Risk of bias in studies	18	Risk of bias for each included study	Not assessed; CLAIM 2024 reporting completeness in Section 3.7 is not a risk-of-bias assessment
Results of individual studies	19	Results for each study	Extraction records in the public repository (Data availability statement)
Results of syntheses	20a	Characteristics and risk of bias of contributing studies	Sections 3.2–3.7

Topic	Item	Checklist item	Location where reported
	20b	Results of all statistical syntheses	Sections 3.2–3.7 (proportions with 95% CIs; Fisher's exact tests)
	20c	Investigations of possible causes of heterogeneity	Sections 3.4.2, 3.6
	20d	Sensitivity analyses	Not performed
Reporting biases	21	Risk of bias due to missing results	Not assessed
Certainty of evidence	22	Certainty in the body of evidence	Not assessed
DISCUSSION			
Discussion	23a	General interpretation in the context of other evidence	Section 4
	23b	Limitations of the evidence	Sections 3.6, 4.3
	23c	Limitations of the review processes	Section 4.3
	23d	Implications for practice, policy and future research	Sections 3.8, 3.9, 4.1, 4.4
OTHER INFORMATION			
Registration and protocol	24a	Registration information, or statement that not registered	Abstract; Section 2.1 (retrospectively registered on OSF Registries, doi:10.17605/OSF.IO/6XJK7)
	24b	Where the protocol can be accessed, or that none was prepared	The OSF registration (doi:10.17605/OSF.IO/6XJK7) documents the review methods; no separate protocol was published
	24c	Amendments to registration or protocol	None
Support	25	Sources of financial or non-financial support	Funding statement
Competing interests	26	Declaration of competing interests	Declaration of competing interest
Availability of data, code and other materials	27	Which materials are publicly available and where	Data availability statement; public repository

Checklist items paraphrased from Page et al., BMJ 2021;372:n71 (PRISMA 2020).

Table S3: Reconstructed topic model and its correspondence to the research gaps. **This table is back-calculated.** The original LDA outputs that informed the gap definitions were lost during a data migration, so we reran the procedure of Section 2.4 on the final corpus of 115 studies, using the limitation and future-work statements recorded for each study during extraction (vocabulary after filtering: 136 terms; terms in fewer than 5% or more than 50% of documents and generic words were removed). C_v coherence for K = 5–20 was 5: 0.279, 6: 0.273, 7: 0.272, 8: 0.257, 9: 0.281, 10: 0.291, 11: 0.300, 12: 0.269, 13: 0.281, 14: 0.299, 15: 0.285, 16: 0.303, 17: 0.318, 18: 0.294, 19: 0.293, 20: 0.319; the highest value selected K = 20. The original run would have produced a table of this form, but its topics and terms may differ. Representative studies are the three with the highest weight on the topic (record IDs as in the public repository); n is the number of studies for which the topic is dominant. The gap column lists up to two gaps whose coding correlates with the topic weight across studies ($r \geq 0.15$). The largest correlation is 0.34, so the reconstructed topics align only loosely with G1–G12: the gaps were defined by reviewer interpretation of the stated limitations (Section 2.4), with topic modeling as an aid rather than the source.

Topic	**Top 10 terms**	**Representative studies**	***n***	**Correlated gaps**
T1	stenosis, lesion, detection, localization, prospective, occlusion, annotation, distal, imaging, training	R2992, R0521, R0514	14	G12 LLMs and reporting ($r = 0.19$); G1 public benchmarks ($r = 0.18$)
T2	frame, segmentation, trained, artifact, detection, quality, vascular, external, vessel, selection	R0546, R1317, R1238	5	none ($r < 0.15$)
T3	expand, occlusion, robustness, augmentation, single, protocol, synthetic, incorporate, segment, frame	R1329, R0550, R2022	4	G4 2D constraints ($r = 0.21$)
T4	aneurysm, training, available, prediction, threshold, system, occlusion, input, segmentation, sequence	R4482, R3155, R1421	5	G7 explainability ($r = 0.34$)
T5	aneurysm, detection, extend, dsa, segmentation, small, frame, extraction, optimize, computational	R4479, R2855, R2114	6	none ($r < 0.15$)
T6	stenosis, classification, larger, video, error, automated, evaluated, end-to-end, view, distal	R4204, R2087, R0099	6	G12 LLMs and reporting ($r = 0.30$); G3 temporal modeling ($r = 0.21$)
T7	segmentation, parameter, temporal, dsa, motion, flow, robustness, end-to-end, frame, learning	R1537, R3999, R1209	6	G7 explainability ($r = 0.16$); G4 2D constraints ($r = 0.15$)
T8	device, region, training, segmentation, computational, frame, validation, larger, dsa, artifact	R1541, R4557, R3576	9	G8 multimodal integration ($r = 0.26$)
T9	cannot, frame, extraction, integrate, segmentation, lesion, gpu, device, combine, dsa	R1251, R1504, R3392	3	none ($r < 0.15$)
T10	frame, artifact, view, time, angiogram, sample, add, noise, external, integration	R1565, R2845, R2377	5	G8 multimodal integration ($r = 0.15$)
T11	aneurysm, motion, complex, flow, method, detection, region, augmentation, value, larger	R2263, R4441, R1862	6	none ($r < 0.15$)
T12	training, real-time, size, inference, automatic, manual, evaluation, dsa, external validation, pressure	R0640, R4143, R0939	6	G6 prospective validation ($r = 0.21$); G1 public benchmarks ($r = 0.19$)
T13	angiography, detection, function, validation, single, coronary artery, prospective, bias, reduce, projection	R0309, R2873, R4208	5	G9 class imbalance ($r = 0.19$)
T14	motion, vessel, segmentation, attention, reduce, frame, small, large, artifact, selection	R4339, R3485, R3910	6	G6 prospective validation ($r = 0.21$)
T15	angiogram, cardiac, coronary artery, guidewire, attention, reference, standard, validation, included, angiography	R4102, R3648, R2466	7	none ($r < 0.15$)
T16	catheter, system, device, contrast, dsa, flow, case, coronary, pressure, small	R2376, R0774, R1422	8	G11 outcome prediction ($r = 0.18$); G6 prospective validation ($r = 0.15$)
T17	case, vessel, structure, projection, large, included, video, excluded, sample, vascular	R3970, R3338, R1222	4	none ($r < 0.15$)
T18	multiple, sequence, ground truth, noise, cardiac, validate, catheter, frame, motion, score	R4411, R4323, R4265	4	G4 2D constraints ($r = 0.21$)

Topic	Top 10 terms	Representative studies	n	Correlated gaps
T19	vessel, segmentation, motion, detection, coronary, coronary artery, trained, occlusion, quality, poor	R2675, R4220, R4699	3	none ($r < 0.15$)
T20	vessel, background, segmentation, structure, module, vascular, gpu, temporal, video, angiography	R0352, R0529, R4762	3	G3 temporal modeling ($r = 0.15$)